RLC3100

High-Precision Smart LCR Tweezers

User Manual • English Version V1



200kHz High-Frequency Measurement | 35-Point Frequency Sweep | AUTO Component Identification

LCR / DIODE / DCV / LED

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1. Product Overview

The RLC3100 is a portable, high-precision smart LCR tweezer meter integrating resistance, capacitance, inductance, diode and DC voltage measurement, LED testing, and component frequency-response scanning.

The instrument supports six test frequencies: 100Hz, 120Hz, 1kHz, 10kHz, 100kHz and 200kHz, with selectable test levels of 0.1V, 0.3V and 0.6Vrms. Its maximum 200kHz test frequency expands its usefulness for certain small capacitors, small inductors and high-frequency components.

AUTO mode identifies resistors, capacitors, inductors, DC voltage and diodes automatically. Series, Parallel and Both equivalent models are supported. Multi-parameter display and 35-point frequency scanning make the instrument useful for electronic design, troubleshooting, component selection and educational experiments.

- High-frequency measurement up to 200kHz
- 35-point component frequency sweep from 100Hz to 200kHz
- AUTO identification of R / C / L / DCV / DIODE
- Simultaneous display of the main and multiple secondary parameters
- LED illumination and polarity identification
- Automatic display rotation for left- or right-handed use

2. Main Specifications and Accuracy

Item	Specification
Model	RLC3100
Display	1.47-inch IPS, 320 × 172
Test Frequency	100Hz / 120Hz / 1kHz / 10kHz / 100kHz / 200kHz
Test Level	0.1V / 0.3V / 0.6Vrms
Test Level at 200kHz	0.1V / 0.3V
DC Voltage Measurement	-32V ~ +32V DC
Frequency Sweep	100Hz – 200kHz, 35 points
Battery Capacity	400mAh

Typical Operating Time	Approximately 4 hours
Charging Port	Type-C
Dimensions	148 × 26 × 14mm

Note Measurement accuracy depends on test frequency, excitation level, component characteristics, tip contact and calibration status.

TYPE	RANGE	100Hz/120Hz	1KHz	10KHz	100KHz	200KHz
RESISTOR	1mR-1R	3%±5	3%±5	3%±5	3%±5	——
	1R-1K	0.5%±5	0.5%±5	0.5%±5	0.5%±5	1%±5
	1K-100K	0.5%±5	0.5%±5	0.5%±5	0.5%±5	2%±5
	100K-500K	0.5%±5	0.5%±5	0.5%±5	1%±5	——
	500k-1M	1%±5	1%±5	1%±5	2%±5	——
	1M-10M	5%±5	5%±5	——	——	——
CAPACITOR	1pF-1nF	——	2%±5	2%±5	1%±5	1%±5
	1nF-1uF	——	1%±5	0.5%±5	0.5%±5	0.5%±5
	1uF-1mF	2%±5	2%±5	2%±5	——	——
	1mF-100mF	3%±5	3%±5	——	——	——
INDUCTOR	50nH-100nH	——	——	——	——	20%±5
	100nH-1uH	——	——	——	10%±5	5%±5
	1uH-10uH	——	5%±5	2%±5	2%±5	1%±5
	10uH-1mH	——	0.5%±5	0.5%±5	0.5%±5	0.5%±5
	1mH-1H	1%±5	1%±5	3%±5	——	——
	1H-10H	1%±5	1%±5	——	——	——

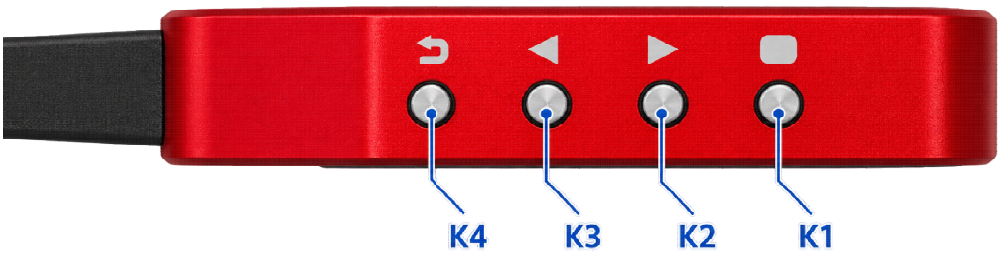
3. Important Safety and Usage Precautions

Important Safety Notice Read and follow these requirements before use.

- Never test energized equipment or components in resistance, capacitance, inductance, diode, LED or any other non-voltage mode.
- Always fully discharge capacitors before measurement.
- Voltage mode is for DC voltage measurement only.
- DC input must not exceed ±32V DC.
- Do not measure AC voltage.
- Do not measure voltage or other components while a USB cable is connected.
- Never connect the test tips directly to mains power, high-voltage supplies or unknown high-energy circuits.
- Verify that the correct operating mode is selected before measurement.

- Keep the tips clean; maintain a consistent grip when testing small capacitors or inductors.
- Avoid touching test nodes when measuring high resistance.
- Avoid high humidity, strong electrostatic fields and corrosive environments during precision measurements.

4. Button Functions and Basic Navigation



Side View of the Four Buttons

Button	Main Function	Common Operation
K1	Confirm / Power / Hold	Short press: confirm selection or hold display Long press: power on/off
K2	Down / Level / Sweep	Short press: move down in menus or change test level Long press: tare in manual capacitance mode
K3	Up / Frequency / Tare	Short press: move up in menus or change test frequency Long press: enter frequency sweep
K4	Mode / Menu / Back	Short press: switch mode, exit or return Long press: enter menu

Tip Button functions may vary with the current screen. Follow the on-screen prompts.

5. Quick Start

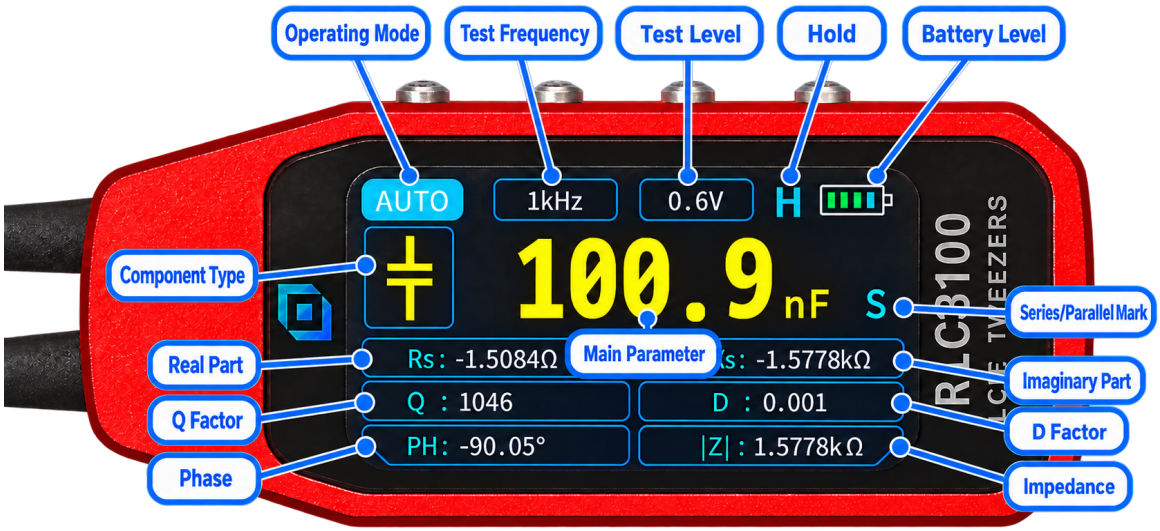
- Press K1 to power on and wait for the main measurement screen.
- Grip the component securely between the two gold-plated test tips.
- In AUTO mode, the instrument identifies the component and displays its main and secondary parameters.
- Press K3 to change test frequency; press K2 to change test level.
- Press K4 to switch measurement modes.



Handheld Measurement Example

Tip For precision, small-capacitance, small-inductance or high-resistance measurements, perform the appropriate calibration first, and keep the tips clean with stable contact.

6. Main Measurement Screen



Main Measurement Screen Example

Description:

1. Operating mode: AUTO, resistance, capacitance, inductance, diode, voltage and LED, in sequence.

2. Excitation frequency: 100Hz, 120Hz, 1kHz, 10kHz, 100kHz, 200kHz.

3. Excitation level: 0.1V, 0.3V, 0.6V; at 200kHz, only 0.1V and 0.3V are supported.

4. Battery level: indicates the instrument battery charge.

5. Component type: corresponds to the selected operating mode.

6. The main parameter is shown in large digits.

7. Series/parallel indicator: S denotes the series model; P denotes the parallel model.

8. Real and imaginary parts: complex measurement data at the current excitation frequency and level.

9. Q and D: the component quality factor and dissipation factor at the current frequency and level.

10. Phase and impedance: the measured phase angle and total impedance at the current frequency and level.

7. AUTO Identification

AUTO mode identifies the component from the measurement results and switches to the appropriate display. It supports automatic identification of resistance, capacitance, inductance, DC voltage and diodes.

Tip For high-loss components, unusual structures or components near identification thresholds, switch to manual R / C / L mode.

AUTO supports five measurement types: resistance, capacitance, inductance, diode and voltage. Its identification algorithm meets routine measurement needs.
If a component falls outside the AUTO identification range, use the corresponding manual mode for more stable results.

8. Manual R / C / L Modes

- Manual R: for resistance and impedance measurements; displays Rs/Rp, Xs/Xp, |Z|, PH and other values.
- Manual C: for capacitance and loss measurements; displays Cs/Cp, Rs, D, |Z|, PH and other values.
- Manual L: for inductance and quality-factor measurements; displays Ls/Lp, Rs, Q, |Z|, PH and other values.

9. Manual Capacitance Tare / Zero

In manual C mode, the tare function temporarily compensates for parasitic capacitance of the test tips and the current test setup.

- Enter manual C mode.
- Do not grip a component yet.
- Gently press the black nylon tweezer arms together so the tip spacing is as close as possible to the spacing used when gripping a component.
- Keep the two tips electrically open.
- Press and hold K2 for approximately 800ms to perform tare / zero.
- Then grip the capacitor to be measured.

Important Tare applies only to the current manual C mode and is stored in RAM. Switching function or mode, changing frequency, briefly pressing the level key, entering calibration or powering off clears the tare data.

10. Test Frequency and Test Level

Test frequencies: 100Hz / 120Hz / 1kHz / 10kHz / 100kHz / 200kHz.

Test levels: 0.1V / 0.3V / 0.6Vrms; 200kHz supports only 0.1V / 0.3V.

Frequency	Recommended Use
100Hz / 120Hz	Large capacitors and low-frequency characterization
1kHz	General RLC measurement
10kHz	Small/medium components and higher-frequency characterization
100kHz	Small capacitors, small inductors and high-frequency components
200kHz	Very small capacitors, small inductors and high-frequency analysis

Tip A component may have different measured parameters at different frequencies. Prefer the frequency and level specified in its datasheet.

Selecting the Excitation Level:

0.1V: For in-circuit measurements. A lower level reduces the influence of other connected devices, such as PN junctions, but may reduce measurement accuracy.

0.6V: The highest excitation level, generally providing the best accuracy. It is used for most measurements.

0.3V: An intermediate level; choose according to the application.

Selecting the Excitation Frequency:

Select frequency according to the component design and actual operating conditions. An arbitrary frequency will not necessarily give the best result. The following choices are generally recommended.

1. 100Hz / 120Hz: for larger capacitors, for example above 10μF.
2. 1kHz: suitable for most components, including resistors, capacitors from 1nF to 10μF, and inductors above 10μH.
3. 10kHz: another common frequency for resistors below 100kΩ, capacitors below

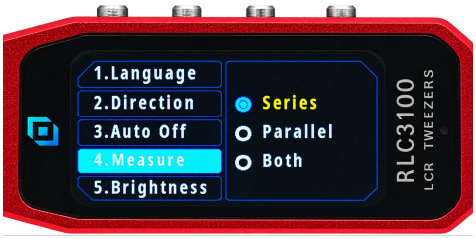
1μF and inductors above 4.7μH.

4. 100kHz / 200kHz: extensions of the 10kHz range for pF-level capacitors and inductors of a few hundred nH. At 200kHz, some wire-wound inductors can be measured down to about 100nH, but ordinary nH-scale inductors may require higher frequencies and better test conditions. These frequencies are also unsuitable for high-value resistors above 100kΩ because parallel parasitic capacitance has a significant effect.

11. Series / Parallel / Both Equivalent Models

The Series model expresses a component using series-equivalent parameters; the Parallel model uses parallel-equivalent parameters. Both mode displays series- and parallel-related results together for comparison.

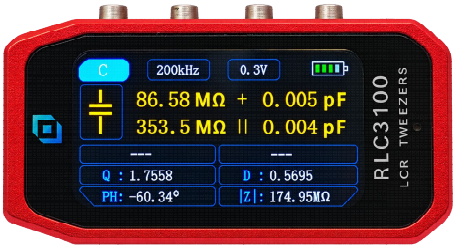
1. Select the measurement model from the menu shown. The Series model is sufficient for most routine measurements.



2. When Series is selected, a cyan “S” appears at the right of the main measurement area, as shown below. When Parallel is selected, “P” is displayed instead.



3. As shown below, Both mode displays series and parallel measurement results simultaneously.



12. Resistance Measurement

Use AUTO or manual R mode. Grip both ends of the resistor, wait for a stable reading, then read the main and secondary parameters. Adjust the frequency, level or equivalent model as needed.

Tip High-resistance measurements are sensitive to humidity, surface leakage, body contact and environmental parasitics. Keep the tips and component surfaces clean and dry.

13. Capacitance Measurement

Use AUTO or manual C mode. Always fully discharge the capacitor before testing. Larger capacitors are generally better measured at lower frequencies, while smaller capacitors can often be measured at higher frequencies.

Warning Make sure the capacitor is fully discharged before measurement. Never directly test a capacitor that still holds charge.

14. Inductance Measurement

Use AUTO or manual L mode. For small inductors, first try 100kHz or 200kHz. For larger inductors, choose a lower frequency according to the component specifications and actual impedance.

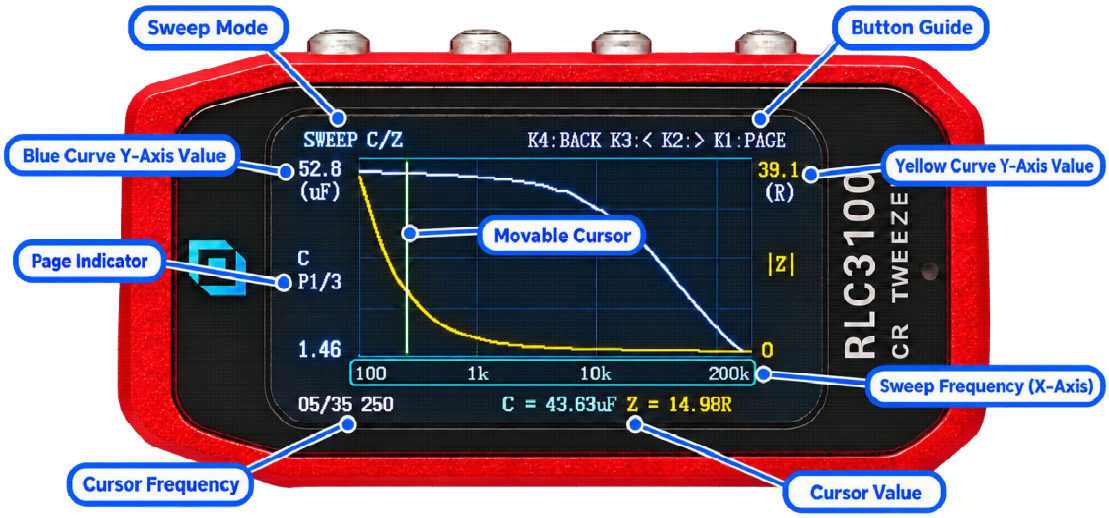
Note Under suitable conditions, some low-parasitic, low-impedance wire-wound inductors can be measured down to approximately 50nH. Not all SMD or other inductor structures can reach the same lower limit.

15. Component Frequency Sweep

The RLC3100 performs an automatic 35-point sweep from 100Hz to 200kHz. The sweep uses a fixed 0.3V test level; no parameter or separate test-level selection is required.

The sweep screen is explained in the following illustration:

Press K4 to exit; K3 and K2 move the cursor left and right; K1 advances to the next page.



Procedure:

On the normal measurement screen, press and hold K3 to enter Sweep → grip the component → press K1 to start scanning → keep the grip steady → wait until all 35 points are measured → view the curves.

Sweep Result Pages:

Inductance and capacitance each have three curve pages; resistance has two. These pages show the main parameters and frequency-dependent trends from different perspectives. Refer to the on-screen labels for curve names and vertical-axis quantities.

After scanning, use K2 and K3 to move the cursor between frequency points and inspect the corresponding values. Press K1 to switch pages; press K4 to exit.

Explanation:

The 1nF capacitor sweep shown above helps illustrate why a test frequency above 1kHz is needed for small capacitors: according to this example, at frequencies below 1kHz the reactance of a 1nF capacitor is too small to measure.

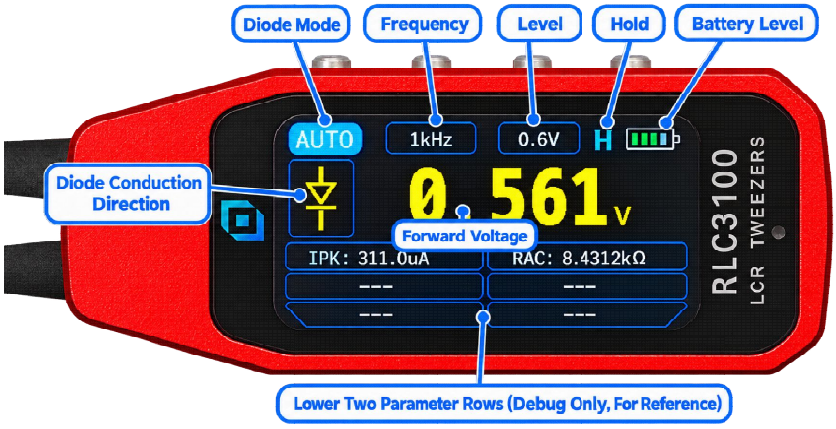
Caution Do not move or re-grip the component during a sweep. Unstable contact can produce local curve anomalies.

16. Diode Testing

Diode mode checks conduction, measures forward voltage drop and identifies conduction direction.

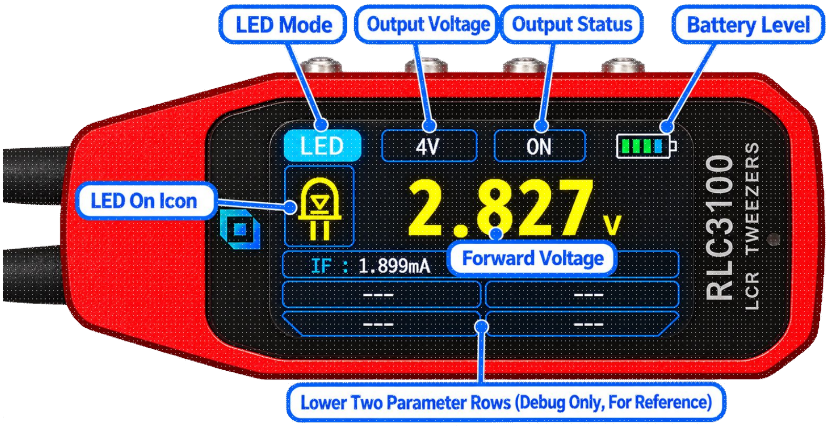
In AUTO mode, only diode forward voltage and conduction direction are identified. Manual D mode also includes a continuity test.

1. Diode test: measures forward voltage drop and identifies conduction direction.
2. Continuity test: a buzzer sounds at low resistance, approximately below 30Ω.



17. LED Testing / Illumination

LED mode provides an approximately 4V test output to check conduction, identify polarity and illuminate common LEDs, including red, yellow, orange, green, blue, white and violet types.

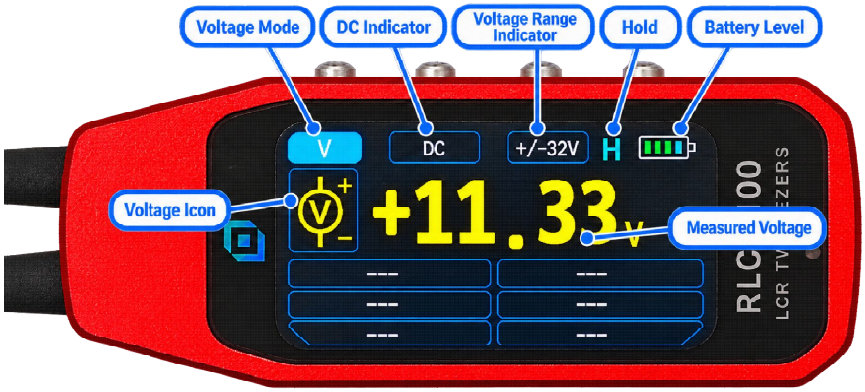


Note Actual LED brightness depends on forward voltage, internal structure and

test conditions. This function is primarily for quickly checking LED conduction and illumination.

18. DC Voltage Measurement

The RLC3100 measures DC voltages from -32V to +32V and automatically displays the polarity.



Warning DC voltage only. Do not measure AC voltage. Input must not exceed ±32V DC. Do not use on mains, high-voltage or unknown high-energy circuits.

19. Automatic Left-/Right-Hand Orientation

The RLC3100 has a built-in orientation sensor that automatically rotates the display according to how it is held, making left- and right-handed operation more convenient. The menu also allows the display to be fixed for left- or right-handed use.



20. System Menu

Press K4 on the main measurement screen to open the system menu. The available items are listed below.

Use K2 / K3 to move the selection, K1 to confirm and K4 to return.



- 1. Language: switch between Chinese and English.
- 2. Direction: set left- or right-handed display orientation.
- 3. Auto Off: select Never, 5 minutes, 10 minutes or 15 minutes.
- 4. Measure: choose Series, Parallel or Both equivalent models.
- 5. Brightness: adjust display brightness.
- 6. Volume: adjust buzzer volume.

- 7. Boot Logo: enable or disable the startup logo.
- 8. Calibration: user-level system calibration (Note: normally do not change calibration items).
- 9. Parameter Memory: choose whether to remember the settings used before power-off.
- 10. Firmware Upgrade: switch to firmware update mode.
- 11. Factory Reset: restore all parameters and calibration data to factory settings.
- 12. Device Information: view system information.

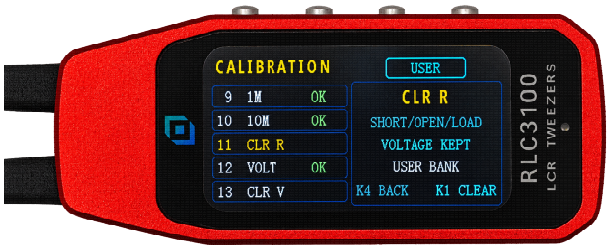
21. User Calibration

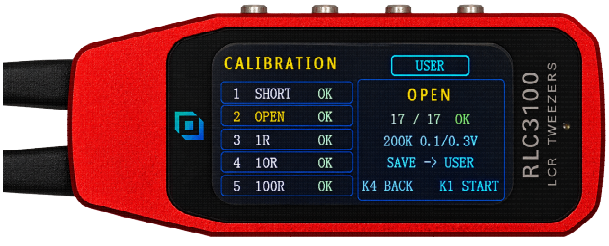
User calibration compensates for errors introduced by the test tips, test setup and measurement channels. Recalibration is recommended for precision measurements, after replacing tips, when the environment changes significantly, or when abnormal open/short residuals are observed.

Caution Avoid interference during calibration. Do not move the tweezers or touch the test nodes. High-resistance calibration is particularly sensitive to humidity, contamination and body coupling.

If calibration settings are changed incorrectly, use Factory Reset to restore the original factory calibration.

- 1. SHORT: short-circuit calibration. Connect the two test tips securely together.
- 2. OPEN: open-circuit calibration. Separate the two tips completely and keep them stable.
- 3. LOAD: standard-resistor calibration. Grip the specified standard resistor (0.1% accuracy required): 1Ω, 10Ω, 100Ω, 1kΩ, 10kΩ, 100kΩ, 1MΩ, 10MΩ.
- 4. Clear resistance calibration: select “11 CLR R” to clear the resistance-related calibration items above.
- 5. Voltage calibration: select “12 VOLT” and follow the prompts to calibrate at +10V and -10V once each (a high-precision 10V voltage source is required).
- 6. Clear voltage calibration: select “13 CLR V” to clear the voltage calibration data above.





22. Restore Factory Calibration

The factory reset function restores user calibration data from the factory calibration data stored in the device. The previous user calibration data will be overwritten.

23. Firmware Upgrade

The RLC3100 supports firmware updates via USB. Obtain firmware update files specifically intended for the RLC3100 from an official source.
Download the latest firmware link:
https://github.com/liangwork168/RLC3100-Firmware/releases

Upgrade Procedure:

- Enter firmware upgrade mode using menu item “10. Firmware Upgrade” or by powering on while holding K4 + K1.
- After entering USB UPDATE, connect the device to a computer with a Type-C data cable, as illustrated.
- Once the computer recognizes the upgrade drive, copy the firmware file to the USB drive.
- Wait for the update to finish; the device exits update mode and starts the application automatically.
- If an error occurs, press K4 or K1 to restart into the application or re-enter update mode.



Caution Do not disconnect USB during an update. Use only an official update file intended for the RLC3100.